ISLAMIC UNIVERSITY OF TECHNOLOGY

DHAKA, BANGLADESH

ORGANIZATION OF ISLAMIC COOPERATION

DATABASE MANAGEMENT SYSTEMS LAB

Name: Antara Arifa Mullick

Student ID: 210042132

Department: Computer Science and Engineering

Program: B.Sc. in Software Engineering

Report preparing for Lab Task

Task 1:

Write SQL statements to create the following tables with the given specifications:

a) DOCTOR:

NAME	VARCHAR2(20)	(e.g.: MR. X, MR. Y)
SPECIALIZATION	CHAR(2)	(e.g. CS, GS, IM, ER)
FEE	NUMBER	(e.g. 2000, 1500)
		Primary Key(NAME, SPECIALIZATION)

Solution Approach:

```
CREATE TABLE DOCTOR
(
NAME VARCHAR2 (20) ,
SPECIALIZATION CHAR (2) ,
FEE NUMBER ,
CONSTRAINT PK_DOCTOR PRIMARY KEY ( NAME ,
SPECIALIZATION )
);
```

SQL Command:

```
SQL> CREATE TABLE DOCTOR
  2  (
  3  NAME VARCHAR2 (20) ,
  4  SPECIALIZATION CHAR (2) ,
  5  FEE NUMBER ,
  6  CONSTRAINT PK_DOCTOR PRIMARY KEY ( NAME , SPECIALIZATION )
  7  );

Table created.
```

b) PATIENT:

PATIENT_NO	CHAR(5)	(e.g.: P-101) Primary Key
NAME	VARCHAR2(20)	(e.g.: A, B, C) Not Null
ADDRESS	VARCHAR2(10)	(e.g.: DHK, KHL, etc.)

Solution Approach:

```
CREATE TABLE PATIENT
(
PATIENT_NO CHAR (5),
NAME VARCHAR2 (20) ,
ADDRESS VARCHAR(10),
CONSTRAINT PK_PATIENT PRIMARY KEY ( PATIENT_NO )
);
```

SQL Command:

```
SQL> CREATE TABLE PATIENT
  2  (
  3  PATIENT_NO CHAR (5),
  4  NAME VARCHAR2 (20) ,
  5  ADDRESS VARCHAR(10),
  6  CONSTRAINT PK_PATIENT PRIMARY KEY ( PATIENT_NO )
  7  );

Table created.
```

c) APPOINTMENT:

PATIENT_NO	CHAR(5)	(e.g.: P-101)
NAME	VARCHAR2(20)	(e.g.: MR. X, MR. Y)
SPECIALIZATION	CHAR(2)	(e.g. CS, GS, IM, ER)
		Primary Key(PATIENT_NO, NAME,SPECIALIZATION)

Solution Approach:

```
CREATE TABLE APPOINTMENT
(
PATIENT_NO CHAR (5),
NAME VARCHAR2 (20) ,
SPECIALIZATION CHAR(2),
CONSTRAINT PK_APPOINTMENT PRIMARY KEY ( PATIENT_NO,
NAME, SPECIALIZATION )
);
```

SQL Command:

```
SQL> CREATE TABLE APPOINTMENT
  2  (
  3  PATIENT_NO CHAR (5),
  4  NAME VARCHAR2 (20) ,
  5  SPECIALIZATION CHAR(2),
  6  CONSTRAINT PK_APPOINTMENT PRIMARY KEY ( PATIENT_NO, NAME, SPECIALIZATION )
  7  );

Table created.
```

Task 2:

Write SQL statements to perform the following alteration operations:

(a)Add a new attribute 'APPOINTMENT_DATE' (DATE type) in APPOINTMENT table.

Solution Approach:

```
ALTER TABLE APPOINTMENT
ADD APPOINTMENT_DATE DATE;
```

SQL Command:

```
SQL> ALTER TABLE APPOINTMENT
  2  ADD APPOINTMENT_DATE DATE;

Table altered.
```

(b) Modify the PRIMARY KEY of APPOINTMENT table and add APPOINTMENT_DATE too with

the previous ones.

Solution Approach:

```
ALTER TABLE APPOINTMENT
DROP CONSTRAINT PK_APPOINTMENT;

ALTER TABLE APPOINTMENT
```

```
ADD CONSTRAINT PK_APPOINTMENT  primary key
(PATIENT_NO, NAME, SPECIALIZATION, APPOINTMENT_DATE);
```

SQL Command:

```
SQL> ALTER TABLE APPOINTMENT
  2  DROP CONSTRAINT PK_APPOINTMENT;

Table altered.

SQL> ALTER TABLE APPOINTMENT
  2  ADD CONSTRAINT PK_APPOINTMENT  primary key (PATIENT_NO, NAME, SPECIALIZATION, APPOINTMENT_DATE
);

Table altered.
```

(c) Rename the attribute PATIENT_NO, NAME from APPOINTMENT table to P_NO and D_NAME

respectively.

Solution Approach:

```
ALTER TABLE APPOINTMENT
RENAME COLUMN PATIENT_NO TO P_NO;

ALTER TABLE APPOINTMENT
RENAME COLUMN NAME TO D_NAME;
```

SQL Command:

```
SQL> ALTER TABLE APPOINTMENT
  2  RENAME COLUMN PATIENT_NO TO P_NO;

Table altered.

SQL> ALTER TABLE APPOINTMENT
  2  RENAME COLUMN NAME TO D_NAME;

Table altered.
```

(d) Rename the table APPOINTMENT to APPOINTMENT_INFO.

Solution Approach:

```
ALTER TABLE APPOINTMENT
```

```
RENAME TO APPOINTMENT_INFO;
```

SQL Command:

```
SQL> ALTER TABLE APPOINTMENT
  2  RENAME TO APPOINTMENT_INFO;

Table altered.
```

(e) Add two foreign key constraints FK_APPOINTMENT_DOCTOR and FK_APPOINTMENT_PATIENT

that identifies D_NAME,SPECIALIZATION and P_NO as foreign keys.

Solution Approach:

```
ALTER TABLE APPOINTMENT_INFO
ADD CONSTRAINT FK_APPOINTMENT_DOCTOR
FOREIGN KEY (D_NAME, SPECIALIZATION) REFERENCES
DOCTOR(NAME, SPECIALIZATION);

ALTER TABLE APPOINTMENT_INFO
ADD CONSTRAINT FK_APPOINTMENT_PATIENT
FOREIGN KEY (P_NO) REFERENCES PATIENT(PATIENT_NO);
```

SQL Command:

```
SQL> ALTER TABLE APPOINTMENT_INFO
  2  ADD CONSTRAINT FK_APPOINTMENT_DOCTOR
  3  FOREIGN KEY (D_NAME, SPECIALIZATION) REFERENCES DOCTOR(NAME, SPECIALIZATION);

Table altered.

SQL>
SQL> ALTER TABLE APPOINTMENT_INFO
  2  ADD CONSTRAINT FK_APPOINTMENT_PATIENT
  3  FOREIGN KEY (P_NO) REFERENCES PATIENT(PATIENT_NO);
```

Task 3:

Insert at least 3 records in each table following the example.

VALUES ('P-103', 'MR. X', 'CS', TO_DATE('2023-08-27', 'YYYY-MM-DD'));

```
INSERT INTO DOCTOR (NAME, SPECIALIZATION, FEE) VALUES
('MR. X', 'CS', 2000);
INSERT INTO DOCTOR (NAME, SPECIALIZATION, FEE) VALUES
('MR. Y', 'GS', 1500);
INSERT INTO DOCTOR (NAME, SPECIALIZATION, FEE) VALUES
('MR. Z', 'GS', 2500);


INSERT INTO PATIENT (PATIENT_NO, NAME, ADDRESS)
VALUES ('P-101', 'A', 'DHK');
INSERT INTO PATIENT (PATIENT_NO, NAME, ADDRESS)
VALUES ('P-102', 'B', 'KHL');
INSERT INTO PATIENT (PATIENT_NO, NAME, ADDRESS)
VALUES ('P-103', 'C', 'DHK');

INSERT INTO APPOINTMENT_INFO (P_NO, D_NAME,
SPECIALIZATION, APPOINTMENT_DATE)
VALUES ('P-101', 'MR. X', 'CS', TO_DATE('2023-08-25',
'YYYY-MM-DD'));
INSERT INTO APPOINTMENT_INFO (P_NO, D_NAME,
SPECIALIZATION, APPOINTMENT_DATE)
VALUES ('P-102', 'MR. Y', 'GS', TO_DATE('2023-08-26',
'YYYY-MM-DD'));
INSERT INTO APPOINTMENT_INFO (P_NO, D_NAME,
SPECIALIZATION, APPOINTMENT_DATE)
VALUES ('P-103', 'MR. X', 'CS', TO_DATE('2023-08-27',
'YYYY-MM-DD'));
```

SQL Command:

```
SQL> INSERT INTO DOCTOR (NAME, SPECIALIZATION, FEE) VALUES ('MR. X', 'CS', 2000);

1 row created.

SQL> INSERT INTO DOCTOR (NAME, SPECIALIZATION, FEE) VALUES ('MR. Y', 'GS', 1500);

1 row created.

SQL> INSERT INTO DOCTOR (NAME, SPECIALIZATION, FEE) VALUES ('MR. Z', 'GS', 2500);

1 row created.

SQL> INSERT INTO PATIENT (PATIENT_NO, NAME, ADDRESS) VALUES ('P-101', 'A', 'DHK');

1 row created.

SQL> INSERT INTO PATIENT (PATIENT_NO, NAME, ADDRESS) VALUES ('P-102', 'B', 'KHL');

1 row created.

SQL> INSERT INTO PATIENT (PATIENT_NO, NAME, ADDRESS) VALUES ('P-103', 'C', 'DHK');

1 row created.

SQL> INSERT INTO APPOINTMENT_INFO (P_NO, D_NAME, SPECIALIZATION, APPOINTMENT_DATE)
  2  VALUES ('P-101', 'MR. X', 'CS', TO_DATE('2023-08-25', 'YYYY-MM-DD'));

1 row created.

SQL> INSERT INTO APPOINTMENT_INFO (P_NO, D_NAME, SPECIALIZATION, APPOINTMENT_DATE)
  2  VALUES ('P-102', 'MR. Y', 'GS', TO_DATE('2023-08-26', 'YYYY-MM-DD'));

1 row created.

SQL> INSERT INTO APPOINTMENT_INFO (P_NO, D_NAME, SPECIALIZATION, APPOINTMENT_DATE)
  2  VALUES ('P-103', 'MR. X', 'CS', TO_DATE('2023-08-27', 'YYYY-MM-DD'));

1 row created.
```

Task 4:

Write SQL statements to answer the following queries:

(a)Find all the Doctors' names whose fees are less than 1500.

Solution Approach:

```sql
SELECT NAME FROM DOCTOR WHERE FEE < 1500;
```

SQL Command:

```
SQL> SELECT NAME FROM DOCTOR WHERE FEE < 1500;

no rows selected
```

(b)Find all the Patients' names who live in 'KHL' city.

Solution Approach:

```
SELECT NAME FROM PATIENT WHERE ADDRESS = 'KHL';
```

SQL Command:

```
SQL> SELECT NAME FROM PATIENT WHERE ADDRESS = 'KHL';

NAME
--------------------
B
```

(c)Show the result of Cartesian Product between PATIENT and APPOINTMENT_INFO table.

Solution Approach:

```
SELECT * FROM PATIENT, APPOINTMENT_INFO;
```

SQL Command:

```
SQL> SELECT * FROM PATIENT, APPOINTMENT_INFO;

PATIE NAME                 ADDRESS    P_NO  D_NAME               SP APPOINTME
----- -------------------- ---------- ----- -------------------- -- ---------
P-101 A                    DHK        P-101 MR. X                CS 25-AUG-23
P-102 B                    KHL        P-101 MR. X                CS 25-AUG-23
P-103 C                    DHK        P-101 MR. X                CS 25-AUG-23
P-101 A                    DHK        P-102 MR. Y                GS 26-AUG-23
P-102 B                    KHL        P-102 MR. Y                GS 26-AUG-23
P-103 C                    DHK        P-102 MR. Y                GS 26-AUG-23
P-101 A                    DHK        P-103 MR. X                CS 27-AUG-23
P-102 B                    KHL        P-103 MR. X                CS 27-AUG-23
P-103 C                    DHK        P-103 MR. X                CS 27-AUG-23

9 rows selected.
```

(d)Show the result of Natural Join between PATIENT and APPOINTMENT_INFO table.

Solution Approach:

```
SELECT * FROM PATIENT
NATURAL JOIN APPOINTMENT_INFO;
```

SQL Command:

```
SQL> SELECT * FROM PATIENT
  2  NATURAL JOIN APPOINTMENT_INFO;

PATIE NAME                 ADDRESS    P_NO  D_NAME               SP APPOINTME
----- -------------------- ---------- ----- -------------------- -- ---------
P-101 A                    DHK        P-101 MR. X                CS 25-AUG-23
P-102 B                    KHL        P-101 MR. X                CS 25-AUG-23
P-103 C                    DHK        P-101 MR. X                CS 25-AUG-23
P-101 A                    DHK        P-102 MR. Y                GS 26-AUG-23
P-102 B                    KHL        P-102 MR. Y                GS 26-AUG-23
P-103 C                    DHK        P-102 MR. Y                GS 26-AUG-23
P-101 A                    DHK        P-103 MR. X                CS 27-AUG-23
P-102 B                    KHL        P-103 MR. X                CS 27-AUG-23
P-103 C                    DHK        P-103 MR. X                CS 27-AUG-23

9 rows selected.
```

(e)Find all the Patient's names and their address who have an appointment today.

Solution Approach:

```
SELECT P.NAME, P.ADDRESS
 FROM PATIENT P
 JOIN APPOINTMENT_INFO A ON P.PATIENT_NO = A.P_NO
 WHERE A.APPOINTMENT_DATE = TRUNC(SYSDATE);
```

SQL Command:

```
SQL> SELECT P.NAME, P.ADDRESS
  2  FROM PATIENT P
  3  JOIN APPOINTMENT_INFO A ON P.PATIENT_NO = A.P_NO
  4  WHERE A.APPOINTMENT_DATE = TRUNC(SYSDATE);

no rows selected
```

(f)Find all the Doctor-related information who have patients from ’DHK’.

Solution Approach:

```
SELECT DISTINCT D.*
 FROM DOCTOR D
 JOIN APPOINTMENT_INFO A ON D.NAME = A.D_NAME AND
D.SPECIALIZATION = A.SPECIALIZATION
 JOIN PATIENT P ON A.P_NO = P.PATIENT_NO
 WHERE P.ADDRESS = 'DHK';
```

SQL Command:

```
SQL> SELECT DISTINCT D.*
  2  FROM DOCTOR D
  3  JOIN APPOINTMENT_INFO A ON D.NAME = A.D_NAME AND D.SPECIALIZATION = A.SPECIALIZATION
  4  JOIN PATIENT P ON A.P_NO = P.PATIENT_NO
  5  WHERE P.ADDRESS = 'DHK';

NAME                 SP        FEE
-------------------- -- ----------
MR. X                CS       2000
```

(g) Find all Patient-related information who has an appointment with a doctor of ’GS’ spe-

cialization or a doctor whose fee is greater than 1500.

Solution Approach:

```
 SELECT DISTINCT P.*
  FROM PATIENT P
  JOIN APPOINTMENT_INFO A ON P.PATIENT_NO = A.P_NO
  JOIN DOCTOR D ON A.D_NAME = D.NAME AND
A.SPECIALIZATION = D.SPECIALIZATION
  WHERE D.SPECIALIZATION = 'GS' OR D.FEE > 1500;
```

SQL Command:

Task 5:

Write following DML statements:

a)Update the NAME and ADDRESS of a tuple from ’A’ and ’DHK ’ to ’K’ and ’RAJ’ accordingly.

Solution Approach:

```
UPDATE PATIENT SET NAME = 'K', ADDRESS = 'RAJ' WHERE
NAME = 'A' AND ADDRESS = 'DHK';
```

SQL Command:

```
SQL> UPDATE PATIENT SET NAME = 'K', ADDRESS = 'RAJ' WHERE NAME = 'A' AND ADDRESS = 'DHK';

1 row updated.
```

a)Update the NAME of table DOCTOR from ’MR. Y’ as ’Ms. Y’.

```
ALTER TABLE APPOINTMENT_INFO
```

```
 DROP CONSTRAINT FK_APPOINTMENT_DOCTOR;

 UPDATE DOCTOR SET NAME = 'Ms. Y' WHERE NAME = 'MR.
Y';

 UPDATE APPOINTMENT_INFO SET D_NAME = 'Ms. Y' WHERE
D_NAME = 'MR. Y';

 ALTER TABLE APPOINTMENT_INFO
 ADD CONSTRAINT FK_APPOINTMENT_DOCTOR
 FOREIGN KEY (D_NAME, SPECIALIZATION) REFERENCES
DOCTOR(NAME, SPECIALIZATION);
```

SQL Command:

```
SQL> ALTER TABLE APPOINTMENT_INFO
  2  DROP CONSTRAINT FK_APPOINTMENT_DOCTOR;

Table altered.

SQL> UPDATE DOCTOR SET NAME = 'Ms. Y' WHERE NAME = 'MR. Y';

1 row updated.

SQL> ALTER TABLE APPOINTMENT_INFO
  2  ADD CONSTRAINT FK_APPOINTMENT_DOCTOR
  3  FOREIGN KEY (D_NAME, SPECIALIZATION) REFERENCES DOCTOR(NAME, SPECIALIZATION);
ADD CONSTRAINT FK_APPOINTMENT_DOCTOR
               *
ERROR at line 2:
ORA-02298: cannot validate (DBMS_170042001.FK_APPOINTMENT_DOCTOR) - parent keys
not found


SQL> UPDATE APPOINTMENT_INFO SET D_NAME = 'Ms. Y' WHERE D_NAME = 'MR. Y';

1 row updated.

SQL> ALTER TABLE APPOINTMENT_INFO
  2  ADD CONSTRAINT FK_APPOINTMENT_DOCTOR
  3  FOREIGN KEY (D_NAME, SPECIALIZATION) REFERENCES DOCTOR(NAME, SPECIALIZATION);

Table altered.
```

c)Delete Patient with PATIENT_NO P-101.

Solution Approach:

```sql
ALTER TABLE APPOINTMENT_INFO
DROP CONSTRAINT FK_APPOINTMENT_PATIENT;

DELETE FROM PATIENT WHERE PATIENT_NO = 'P-101';
```

SQL Command:

```
SQL> ALTER TABLE APPOINTMENT_INFO
  2  DROP CONSTRAINT FK_APPOINTMENT_PATIENT;

Table altered.

SQL> DELETE FROM PATIENT WHERE PATIENT_NO = 'P-101';

1 row deleted.
```

(d)Delete all the information without deleting the table structure.

Solution Approach:

```sql
TRUNCATE TABLE APPOINTMENT_INFO;

 ALTER TABLE APPOINTMENT_INFO
 DISABLE CONSTRAINT FK_APPOINTMENT_DOCTOR;
 TRUNCATE TABLE DOCTOR;

 TRUNCATE TABLE PATIENT;

  INSERT INTO DOCTOR (NAME, SPECIALIZATION, FEE)
VALUES ('MR. X', 'CS', 2000);
  INSERT INTO DOCTOR (NAME, SPECIALIZATION, FEE)
VALUES ('Ms. Y', 'GS', 1500);

  INSERT INTO PATIENT (PATIENT_NO, NAME, ADDRESS)
VALUES ('P-101', 'A', 'DHK');
  INSERT INTO PATIENT (PATIENT_NO, NAME, ADDRESS)
VALUES ('P-102', 'B', 'KHL');
```

```
  INSERT INTO PATIENT (PATIENT_NO, NAME, ADDRESS)
VALUES ('P-103', 'C', 'DHK');

 INSERT INTO APPOINTMENT_INFO (P_NO, D_NAME,
SPECIALIZATION, APPOINTMENT_DATE)
 VALUES ('P-101', 'MR. X', 'CS', TO_DATE('2023-08-
25', 'YYYY-MM-DD'));
 INSERT INTO APPOINTMENT_INFO (P_NO, D_NAME,
SPECIALIZATION, APPOINTMENT_DATE)
 VALUES ('P-102', 'Ms. Y', 'GS', TO_DATE('2023-08-
26', 'YYYY-MM-DD'));
 INSERT INTO APPOINTMENT_INFO (P_NO, D_NAME,
SPECIALIZATION, APPOINTMENT_DATE)
 VALUES ('P-103', 'MR. X', 'CS', TO_DATE('2023-08-
27', 'YYYY-MM-DD'));
```

SQL Command:

```
SQL> TRUNCATE TABLE APPOINTMENT_INFO;

Table truncated.

SQL> TRUNCATE TABLE DOCTOR;
TRUNCATE TABLE DOCTOR
               *
ERROR at line 1:
ORA-02266: unique/primary keys in table referenced by enabled foreign keys


SQL> TRUNCATE TABLE PATIENT;

Table truncated.

SQL> INSERT INTO DOCTOR (NAME, SPECIALIZATION, FEE) VALUES ('MR. X', 'CS', 2000);
INSERT INTO DOCTOR (NAME, SPECIALIZATION, FEE) VALUES ('MR. X', 'CS', 2000)
*
ERROR at line 1:
ORA-00001: unique constraint (DBMS_1700422001.PK_DOCTOR) violated


SQL> INSERT INTO DOCTOR (NAME, SPECIALIZATION, FEE) VALUES ('Ms. Y', 'GS', 1500);
INSERT INTO DOCTOR (NAME, SPECIALIZATION, FEE) VALUES ('Ms. Y', 'GS', 1500)
*
ERROR at line 1:
ORA-00001: unique constraint (DBMS_1700422001.PK_DOCTOR) violated


SQL> INSERT INTO PATIENT (PATIENT_NO, NAME, ADDRESS) VALUES ('P-101', 'A', 'DHK');

1 row created.

SQL> INSERT INTO PATIENT (PATIENT_NO, NAME, ADDRESS) VALUES ('P-102', 'B', 'KHL');

1 row created.

SQL> INSERT INTO PATIENT (PATIENT_NO, NAME, ADDRESS) VALUES ('P-103', 'C', 'DHK');

1 row created.

SQL>
SQL> INSERT INTO APPOINTMENT_INFO (P_NO, D_NAME, SPECIALIZATION, APPOINTMENT_DATE)
  2  VALUES ('P-101', 'MR. X', 'CS', TO_DATE('2023-08-25', 'YYYY-MM-DD'));

1 row created.

SQL> INSERT INTO APPOINTMENT_INFO (P_NO, D_NAME, SPECIALIZATION, APPOINTMENT_DATE)
  2  VALUES ('P-102', 'Ms. Y', 'GS', TO_DATE('2023-08-26', 'YYYY-MM-DD'));

1 row created.

SQL> INSERT INTO APPOINTMENT_INFO (P_NO, D_NAME, SPECIALIZATION, APPOINTMENT_DATE)
  2  VALUES ('P-103', 'MR. X', 'CS', TO_DATE('2023-08-27', 'YYYY-MM-DD'));

1 row created.

SQL> DELETE FROM APPOINTMENT_INFO WHERE D_NAME IN (SELECT NAME FROM DOCTOR);

3 rows deleted.

SQL> TRUNCATE TABLE DOCTOR;
TRUNCATE TABLE DOCTOR
               *
ERROR at line 1:
ORA-02266: unique/primary keys in table referenced by enabled foreign keys


SQL> ALTER TABLE APPOINTMENT_INFO
  2  DISABLE CONSTRAINT FK_APPOINTMENT_DOCTOR;

Table altered.

SQL> TRUNCATE TABLE DOCTOR;

Table truncated.

SQL> INSERT INTO DOCTOR (NAME, SPECIALIZATION, FEE) VALUES ('MR. X', 'CS', 2000);

1 row created.

SQL> INSERT INTO DOCTOR (NAME, SPECIALIZATION, FEE) VALUES ('Ms. Y', 'GS', 1500);

1 row created.

SQL>
```

```
SQL> SELECT DISTINCT P.*
  2  FROM PATIENT P
  3  JOIN APPOINTMENT_INFO A ON P.PATIENT_NO = A.P_NO
  4  JOIN DOCTOR D ON A.D_NAME = D.NAME AND A.SPECIALIZATION = D.SPECIALIZATION
  5  WHERE D.SPECIALIZATION = 'GS' OR D.FEE > 1500;

PATIE NAME                 ADDRESS
----- -------------------- ----------
P-103 C                    DHK
P-102 B                    KHL
P-101 A                    DHK
```

Problems Faced:

1. Syntax Errors in CREATE TABLE Statements:

- Incorrect syntax in the CREATE TABLE statement resulted in table creation failures.

- Solution: Double-checking the syntax against SQL documentation or examples to ensure accuracy.

2. Data Integrity Issues in INSERT Statements:

- Providing incorrect values or omitting values in the INSERT statements resulted in data integrity issues.

- Solution: Thorough validation and accuracy checks when inserting data.

3. Handling NULL Values:

- Attributes with NULL values did not behave as expected in comparisons.

- Solution: Properly handling NULL values in queries and data processing.

4. Query Performance Optimization:

- Poorly optimized queries led to slow performance, especially with large datasets.

- Solution: Efficient use of indexes and optimization of complex queries.

5. Permanent Data Deletion:

- Dropping a table permanently removes its data.

- Solution: Exercise caution when performing such operations to prevent data loss.